

HES Missing Event Investigation Tool — Guide

Revision History

Date	Version	Author	Note
2026-08-12	1.0	Hau Pham	Initial version — full workflow tested end-to-end on 4 real cases

Purpose

Automates the HES "missing event" investigation workflow described in *[20250827] Report - HES Event Evaluation Workflow.docx*: given a study FID (or hash) and the event timestamp a technician reported (Brady/Tachy/Pause), it downloads the right data-all/.anb/report-hourly/device-log files from S3, converts anb->ann, aggregates the hourly event reports, parses the study's real settings straight from the device log, computes whether the event *should* have triggered, cross-checks the device log and report-hourly for what actually happened, shows the ECG waveform for a final visual sanity check, and drafts a ready-to-paste customer-reply write-up.

For how the tool works internally (pipeline steps, assessment logic, known bugs already fixed), see [CLAUDE.md](#) instead — this file is the usage guide.

Prerequisites

- **Python 3.10+** (tested with 3.14 — every module uses `from __future__ import annotations`, no pinned-old-wheel constraint like some sibling tools have).
- AWS SSO access (profile `fw-btcy-sso`), for S3 downloads. Check session validity with:

```
aws sts get-caller-identity --profile fw-btcy-sso
```

A printed account/ARN means the session is valid. If expired:

```
aws sso login --profile fw-btcy-sso
```

- SSH access to the `customer` host alias (resolved via your own `~/.ssh/config`), needed for FID → study hash resolution (issue 524). Check with:

```
ssh customer "echo ok"
```

If this hangs or fails, FID lookups won't work — either fix SSH access, or always pass `--hash`/a full hash directly instead of a FID.

- Run `setup.bat` once (creates `.venv` + installs `requirements.txt`):

```
setup.bat
```

- Launch the GUI:

```
run.bat
```

Or use the CLI directly (for scripted/automated use — see "CLI usage reference" at the end of this guide):

```
.venv\Scripts\python.exe src\main.py --fid 629209 --event-type Pause --date 2026-06-19 --time 19:34:43
```

GUI overview

Two tabs, both backed by the exact same pipeline the CLI calls (on a background thread, so the window stays responsive during S3 downloads). See the worked examples below for the full step-by-step flow.

Input tab:

The screenshot shows the 'Input' tab of the GUI. It contains several input fields and a 'Run' button. Red boxes and numbers 1 through 5 highlight specific areas:

- 1**: Points to the 'Study FID / hash' field, which contains '629209'.
- 2**: Points to the 'Input folder' field, which contains 'w:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031'.
- 3**: Points to the 'Event type' dropdown menu, which is set to 'Tachy'.
- 4**: Points to the 'Run' button.
- 5**: Points to the 'Result' text area, which displays a log of events and a summary of the automated assessment.

The 'Result' text area contains the following text:

```
2026-08-12 14:30:51,374 [INFO] 'w:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031' already has data-all-* files -- reusing them, skipping S3 entirely.
2026-08-12 14:30:51,374 [INFO] Picked data-all file: data-all-05-09-26-12-49-44-16 (tz_offset=-240min) -> event_utc=2026-05-09 21:10:31
2026-08-12 14:30:51,381 [INFO] ann-05-09-26-12-49-44-16.ann already present -- skipping annb->ann conversion.
2026-08-12 14:30:51,382 [INFO] Aggregated hourly events -> w:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031\all_hourly_event.txt
2026-08-12 14:30:51,393 [INFO] Device HW ver: 1.3 FW ver: 1.0.0.40
2026-08-12 14:30:51,496 [INFO] Done. Output folder: w:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031
2026-08-12 14:30:51,499 [INFO] HW ver: 1.3 FW ver: 1.0.0.40
2026-08-12 14:30:51,500 [INFO] StartStudy @ 2026-05-09T16:34:00 UTC0 (from log-daily-05-08-26-20-00-00-16.txt)
FriendlyId = 609493
AlgLead = 2 (middle channel)
Brady/Tachy = 40/145 bpm
PauseThresh = 2500 ms (2.5s)
PauseDetect = enabled
AFibDetect = enabled
Pre/PostEvSecs = 30s / 30s
2026-08-12 14:30:51,500 [INFO] === Automated assessment: Tachy at 2026-05-09 21:10:31 UTC0 (window 2026-05-09 21:09:31 .. 2026-05-09 21:11:31) ===
Consecutive qualifying-beat run CLOSEST TO THE REPORTED EVENT: 2/5 (from 2026-05-09 21:10:39 to 2026-05-09 21:10:40),
=> HES detected too few beats -> not enough beats for 5 consecutive beats -> this Tachy event was missed.
2026-08-12 14:30:51,500 [REPLY] ----- Customer reply draft -----
Study 609493_2026-05-09_171031, event time 5/9 17:10:31
1. General information:
FW version: 1.0.0.40, HW version: 1.3
Study's AlgLead is the middle channel (channel 2), not changed during the study
Tachy threshold is 145bpm
2. Reason for the missed event:
Main cause: HES detected too few beats -> not enough beats exceeding 145bpm for 5 consecutive beats -> this Tachy event was missed.
```

1. Study FID / hash — 6-digit FID or a full study hash

2. **Input folder** — optional; reuses the folder if it already has the data, else downloads into it
3. **Event type / date-time / window** — the reported event, exactly as given (timezone is auto-detected after Run)
4. **Run** — starts the pipeline; progress bar + status show live progress
5. **Result** — full log, ending in the conclusion + a ready-to-paste customer-reply draft (green)

ECG Viewer tab (auto-opens on success, centered on the event):



1. **Toolbar** — window length (3-60s), Jump to event, N/VES/SVES/RR/BPM/ Events show-hide toggles, Export image
2. **Channels** — per-channel show/hide
3. **Seek** — jump to a specific beat number or local time
4. **Chart** — minimap (top) + main waveform, colored beat markers, red dashed line = reported event
5. **Scrollbar** — page through the record

Dependencies

- Python 3.10+, [aws](#) CLI on PATH (profile [fw-btcy-sso](#)), SSH access to the [customer](#) host alias.
- See [requirements.txt](#) ([wfdb](#), [matplotlib](#), [PySide6](#) — PySide6 is only needed for the GUI; the CLI doesn't import it).

Worked examples: 4 real support-email cases, full workflow

Each example below is a **full, real, from-scratch GUI run** for a real "missing event" support case — Study FID only was entered (no shortcut folder), so every run shown here genuinely did: FID → hash resolution over real SSH (issue 524) → real **aws s3** downloads → anb→ann conversion → hourly-event aggregation → settings parsing → the automated assessment → the interactive ECG Viewer.

Each case's previously-downloaded folder was deleted right before its run so the tool couldn't shortcut to "reuse existing data" — every screenshot below is from a genuine live download, not a replay. Each screenshot's numbered markers are explained in its own caption right below it — the numbering restarts at 1 for every image, it does not carry across images.

Example 1 — Study 609493, Tachy @ 2026-05-09 17:10:31

Email: *Study 609493, Tachy event reported at 5/9 17:10:31*

Step 1 — enter the study and event, then click Run

The screenshot shows the ECG Viewer GUI with the following input fields and values:

- Study FID / hash: 609493
- Input folder: (optional -- reuses this folder if it already has the data, else downloads into it)
- Event type: Tachy
- Event date/time (as reported, no conversion needed): 2026-05-09 17:10:31
- Detected timezone: (detected after Run)
- Window (+/- sec): 60

The Run button is highlighted with a red box and a red circle with the number 4. The Open log folder button is also visible.

1. Study FID typed into "Study FID / hash" (Input folder left empty)
2. Event type selected
3. Event date/time entered exactly as reported (local/study clock)
4. **Run** clicked

Step 2 — mid-run (real SSH hash resolution + S3 download in progress)

Input

ECG Viewer

Study FID / hash:

609493

Input folder:

(optional -- reuses this folder if it already has the data, else downloads into it)

Browse...

Event type:

Tachy

▼

Event date/time (as reported, no conversion needed):

2026-05-09

▼

17:10:31

^

▼

Detected timezone:

(detected after Run)

Window (+/- sec):

60

^

▼

Run

Open log folder

1

Progress bar at 95%

2

Running...

3

Result:

```
2026-08-12 14:26:20,617 [INFO] Resolving study hash via get_study_information.py (issue 524)...
2026-08-12 14:26:20,617 [INFO] $ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_609493.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_609493.json
2026-08-12 14:26:51,150 [INFO] study_hash = 69ff61f3367813732cdccdd4
2026-08-12 14:26:52,685 [INFO] Picked data-all file: data-all-05-09-26-12-49-44-16 (tz_offset=-240min) -> event_utc=2026-05-09 21:10:31
2026-08-12 14:27:16,597 [INFO] Running anb2ann_y4_utc0.exe ...
2026-08-12 14:27:16,644 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031\all_hourly_event.txt
2026-08-12 14:27:16,655 [INFO] Device HW ver: 1.3 FW ver: 1.0.0.40
```

1. Progress bar at 95% — Run button is greyed out while busy
2. Status line: "Running..."
3. Live log: hash resolved via the real SSH/npm call, data-all file picked, anb→ann converted, hourly events aggregated, HW/FW version read

Step 3 — final result

Input ECG Viewer

Study FID / hash: 609493

Input folder: (optional -- reuses this folder if it already has the data, else downloads into it) Browse...

Event type: Tachy

Event date/time (as reported, no conversion needed): 2026-05-09 17:10:31

Detected timezone: UTC-4 (detected from data-all-05-09-26-12-49-44-16)

Window (+/- sec): 60

Run Open log folder

100%

ie. Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031

Result:

2026-08-12 14:26:20.617 [INFO] Resolving study hash via get_study_information.py (issue 524)...

2026-08-12 14:26:20.617 [INFO] \$ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event_tmp_fid_609493.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event_tmp_fid_609493.json

2026-08-12 14:26:51.150 [INFO] study hash = 69ff6f3367813732cdcc44

2026-08-12 14:26:52.685 [INFO] Picked data-all file: data-all-05-09-26-12-49-44-16 (tz_offset=-240min) -> event_utc=2026-05-09 21:10:31

2026-08-12 14:27:16.597 [INFO] Running an2ann_v4_utc0.exe ...

2026-08-12 14:27:16.644 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031\all_hourly_event.txt

2026-08-12 14:27:16.655 [INFO] Device HW ver: 1.3 FW ver: 1.0.0.40

2026-08-12 14:27:16.762 [INFO] Done. Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\609493_2026-05-09_171031

2026-08-12 14:27:16.799 [INFO] HW ver: 1.3 FW ver: 1.0.0.40

2026-08-12 14:27:16.800 [INFO] StartStudy @ 2026-05-09T16:34:00 UTC0 (from log-daily-05-08-26-20-00-00-16.txt)

FriendlyId = 609493

AlgLead = 2 (middle channel)

Brady/Tachy = 40/145 bpm

PauseThresh = 2500 ms (2.5s)

PauseDetect = enabled

AFibDetect = enabled

Pre/PostEvSecs = 30s / 30s

2026-08-12 14:27:16.800 [INFO] === Automated assessment: Tachy at 2026-05-09 21:10:31 UTC0 (window 2026-05-09 21:09:31 .. 2026-05-09 21:11:31) ===

Consecutive qualifying-beat run CLOSEST TO THE REPORTED EVENT: 2/5 (from 2026-05-09 21:10:39 to 2026-05-09 21:10:40).

=> HES detected too few beats -> not enough beats for 5 consecutive beats -> this Tachy event was missed.

2026-08-12 14:27:16.800 [REPLY] ----- Customer reply draft -----

Study 609493, event time 5/9 17:10:31

1. General information:

FW version: 1.0.0.40, HW version: 1.3

Study's AlgLead is the middle channel (channel 2), not changed during the study

Tachy threshold is 145bpm

2. Reason for the missed event:

Main cause: HES detected too few beats -> not enough beats exceeding 145bpm for 5 consecutive beats -> this Tachy event was missed.

1. Progress bar reaches 100%
2. Status line: "Done. Output folder: ..."
3. Full log, ending in the automated assessment and the green ready-to-paste customer-reply draft

Step 4 — ECG Viewer (auto-opened, centered on the event)



1. Waveform centered on the reported event, with the beat annotations HES actually produced

Conclusion: only a 2/5 consecutive-beat run near the reported time (21:10:39-21:10:40 UTC0) — not enough to trigger a Tachy event. AlgLead = channel 2 (middle), Tachy threshold 145bpm, FW 1.0.0.40 / HW 1.3.

Example 2 — Study 608418, Tachy @ 2026-05-11 22:28:59

Email: *Study 608418, Tachy event reported at 5/11 22:28:59*

Step 1

The screenshot shows the 'Input' tab of the 'ECG Viewer' application. The form contains the following fields and controls:

- Study FID / hash:** A text input field containing '608418', highlighted with a red box and a red circle with the number 1.
- Input folder:** A text input field with a placeholder '(optional -- reuses this folder if it already has the data, else downloads into it)' and a 'Browse...' button.
- Event type:** A dropdown menu with 'Tachy' selected, highlighted with a red box and a red circle with the number 2.
- Event date/time (as reported, no conversion needed):** A date and time input field containing '2026-05-11' and '22:28:59', highlighted with a red box and a red circle with the number 3.
- Detected timezone:** A text input field containing '(detected after Run)'.
- Window (+/- sec):** A text input field containing '60'.
- Run button:** A button labeled 'Run', highlighted with a red box and a red circle with the number 4.
- Open log folder:** A button labeled 'Open log folder'.
- Result:** A large empty rectangular area for the output.

1. Study FID entered
2. Event type selected
3. Event date/time entered exactly as reported (local/study clock)
4. **Run** clicked

Step 2 — mid-run

Input

ECG Viewer

Study FID / hash:

608418

Input folder:

(optional -- reuses this folder if it already has the data, else downloads into it)

Browse...

Event type:

Tachy

Event date/time (as reported, no conversion needed):

2026-05-11

22:28:59

Detected timezone:

(detected after Run)

Window (+/- sec):

60

Run

Open log folder

1

95%

2

Running...

3

Result:

2026-08-12 14:27:18,544 [INFO] Resolving study hash via get_study_information.py (issue 524)...
2026-08-12 14:27:18,544 [INFO] \$ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event_tmp_fid_608418.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event_tmp_fid_608418.json
2026-08-12 14:27:48,591 [INFO] study_hash = 69fcb0c7367813732cc54643
2026-08-12 14:27:50,174 [INFO] Picked data-all file: data-all-05-11-26-16-13-17-16 (tz_offset=-240min) -> event_utc=2026-05-12 02:28:59
2026-08-12 14:28:18,387 [INFO] Running anb2ann_v4_utc0.exe ...
2026-08-12 14:28:18,502 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\608418_2026-05-11_222859\all_hourly_event.txt
2026-08-12 14:28:18,508 [INFO] Device HW ver: 2.1 FW ver: 1.0.0.40

1. Progress bar mid-run
2. Status: "Running..."
3. Live log of the real SSH/S3 steps as they happen

Step 3 — final result

Input

ECG Viewer

Study FID / hash:

608418

Input folder:

(optional -- reuses this folder if it already has the data, else downloads into it)

Browse...

Event type:

Tachy

Event date/time (as reported, no conversion needed):

2026-05-11

22:28:59

Detected timezone:

UTC-4 (detected from data-all-05-11-26-16-13-17-16)

Window (+/- sec):

60

Run

Open log folder

1

100%

2

Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\608418_2026-05-11_222859

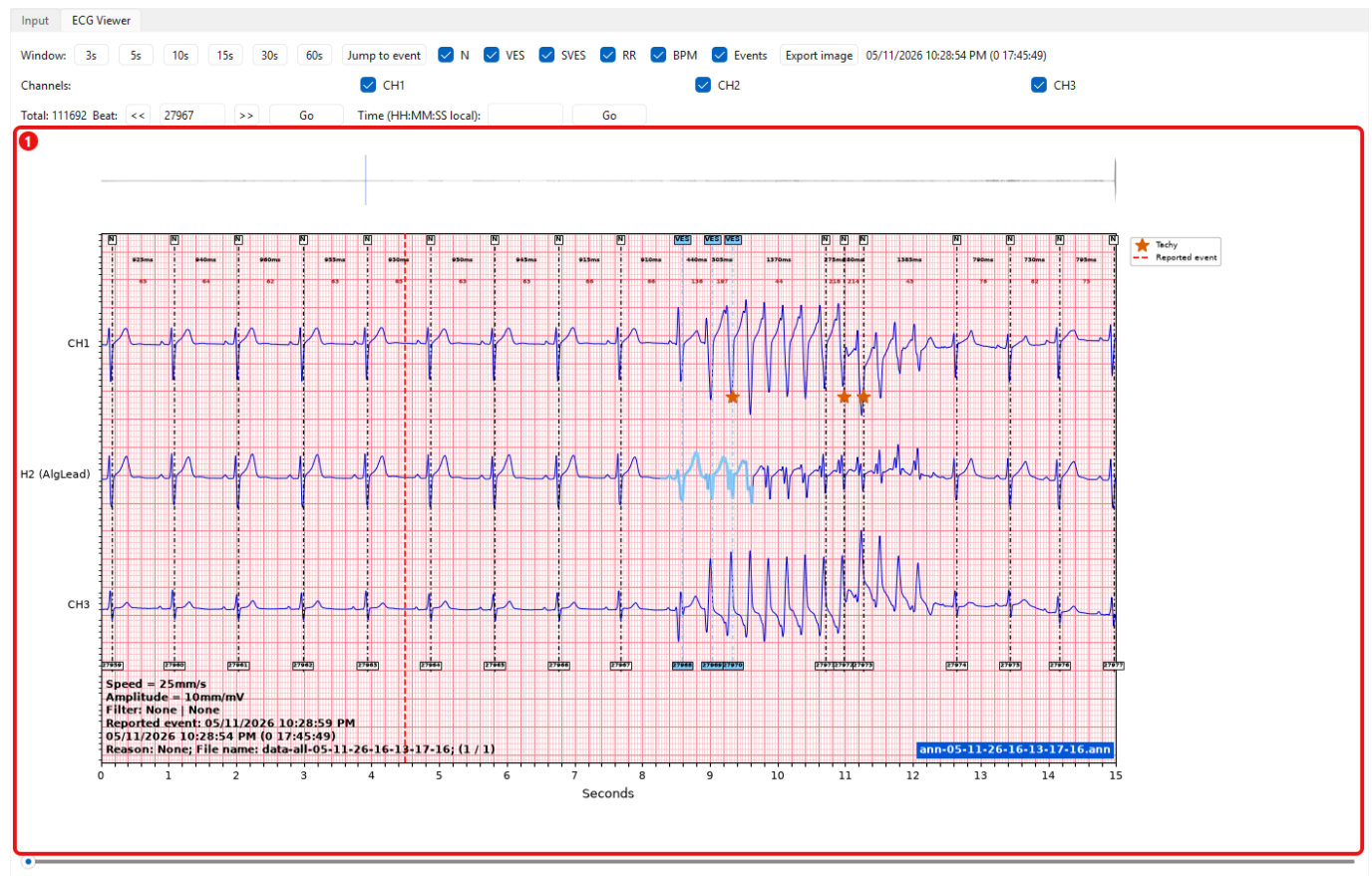
3

Result:

2026-08-12 14:27:18,544 [INFO] Resolving study hash via get_study_information.py (issue 524)...
2026-08-12 14:27:18,544 [INFO] \$ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event_tmp_fid_608418.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event_tmp_fid_608418.json
2026-08-12 14:27:48,591 [INFO] study_hash = 69fcb0c7367813732cc54643
2026-08-12 14:27:50,174 [INFO] Picked data-all file: data-all-05-11-26-16-13-17-16 (tz_offset=-240min) -> event_utc=2026-05-12 02:28:59
2026-08-12 14:28:18,387 [INFO] Running anb2ann_v4_utc0.exe ...
2026-08-12 14:28:18,502 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\608418_2026-05-11_222859\all_hourly_event.txt
2026-08-12 14:28:18,508 [INFO] Device HW ver: 2.1 FW ver: 1.0.0.40
2026-08-12 14:28:18,764 [INFO] Done. Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\608418_2026-05-11_222859
2026-08-12 14:28:18,805 [INFO] HW ver: 2.1 FW ver: 1.0.0.40
2026-08-12 14:28:18,805 [INFO] StartStudy @ 2026-05-09T21:48:34 UTC0 (from log-daily-05-08-26-20-00-00-16.txt)
FriendlyId = 0
AlgLead = 2 (middle channel)
Brady/Tachy = 40/145 bpm
PauseThresh = 2500 ms (2.5s)
PauseDetect = enabled
AFibDetect = enabled
Pre/PostEvSecs = 30s / 30s
2026-08-12 14:28:18,805 [INFO] === Automated assessment: Tachy at 2026-05-12 02:28:59 UTC0 (window 2026-05-12 02:27:59 .. 2026-05-12 02:29:59) ===
Consecutive qualifying-beat run CLOSEST TO THE REPORTED EVENT: 1/5 (from 2026-05-12 02:29:05 to 2026-05-12 02:29:05).
Note: a longer run exists (2/5) from 2026-05-12 02:29:05 to 2026-05-12 02:29:06 in the same window but NOT close to the reported event -> likely an unrelated, separate event.
=> HES detected too few beats -> not enough beats for 5 consecutive beats -> this Tachy event was missed.
2026-08-12 14:28:18,806 [REPLY] ----- Customer reply draft -----
Study 608418, event time 5/11 22:28:59
1. General information:
FW version: 1.0.0.40, HW version: 2.1
Study's AlgLead is the middle channel (channel 2), not changed during the study
Tachy threshold is 145bpm
2. Reason for the missed event:
Main cause: HES detected too few beats -> not enough beats exceeding 145bpm for 5 consecutive beats -> this Tachy event was missed.

1. Progress bar at 100%
2. Status: "Done."
3. Full log and conclusion

Step 4 — ECG Viewer



1. Waveform centered on the reported event

Conclusion: closest run is only 1/5 consecutive qualifying beats (a separate 2/5 run exists nearby but isn't close to the reported time, so it's flagged as a likely unrelated event). AlgLead = channel 2 (middle), Tachy threshold 145bpm, FW 1.0.0.40 / HW 2.1.

Example 3 — Study 629209, Pause @ 2026-06-19 19:34:43

Email: Study 629209, Pause event reported at 6/19 19:34:43

Step 1

The screenshot shows the 'Input' tab of the 'ECG Viewer' application. The form contains the following fields and controls:

- Study FID / hash:** A text input field containing '3209'. A red box with a '1' is around this field.
- Input folder:** A text input field with a 'Browse...' button. A red box with a '2' is around this field.
- Event type:** A dropdown menu showing 'ause'. A red box with a '2' is around this field.
- Event date/time (as reported, no conversion needed):** A date/time input field showing '26-06-19' and '19:34:43'. A red box with a '3' is around this field.
- Detected timezone:** A text input field showing '(detected after Run)'.
- Window (+/- sec):** A text input field showing '60'.
- Run button:** A button labeled 'Run'. A red box with a '4' is around this button.
- Open log folder:** A button labeled 'Open log folder'.
- Progress bar:** A progress bar at the bottom right showing '0%'.
- Result:** A large empty box labeled 'Result:'.

1. Study FID entered
2. Event type selected
3. Event date/time entered exactly as reported (local/study clock)
4. **Run** clicked

Step 2 — mid-run

InputECG Viewer

Study FID / hash:

629209

Input folder:

(optional -- reuses this folder if it already has the data, else downloads into it)

Browse...

Event type:

Pause

Event date/time (as reported, no conversion needed):

2026-06-19

19:34:43

Detected timezone:

(detected after Run)

Window (+/- sec):

60

Run

Open log folder

1

95%

2

Running...

3

Result:

2026-08-12 14:28:21,021 [INFO] Resolving study hash via get_study_information.py (issue 524)...

2026-08-12 14:28:21,021 [INFO] \$ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_629209.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_629209.json

2026-08-12 14:28:53,075 [INFO] study_hash = 6a35a35fe64ae6f085e27d7f

2026-08-12 14:28:54,684 [INFO] Picked data-all file: data-all-06-19-26-13-39-16-28 (tz_offset=-420min) -> event_utc=2026-06-20 02:34:43

2026-08-12 14:29:20,176 [INFO] Running anb2ann_v4_utc0.exe ...

2026-08-12 14:29:20,190 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\629209_2026-06-19_193443\all_hourly_event.txt

2026-08-12 14:29:20,209 [INFO] Device HW ver: 3.1 FW ver: 4.0.014.3

- 1. Progress bar mid-run
- 2. Status: "Running..."
- 3. Live log of the real SSH/S3 steps

Step 3 — final result

InputECG Viewer

Study FID / hash:

629209

Input folder:

(optional -- reuses this folder if it already has the data, else downloads into it)

Browse...

Event type:

Pause

Event date/time (as reported, no conversion needed):

2026-06-19

19:34:43

Detected timezone:

UTC-7 (detected from data-all-06-19-26-13-39-16-28)

Window (+/- sec):

60

Run

Open log folder

1

100%

2

Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\629209_2026-06-19_193443

3

Result:

2026-08-12 14:28:21,021 [INFO] Resolving study hash via get_study_information.py (issue 524)...

2026-08-12 14:28:21,021 [INFO] \$ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_629209.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_629209.json

2026-08-12 14:28:53,075 [INFO] study_hash = 6a35a35fe64ae6f085e27d7f

2026-08-12 14:28:54,684 [INFO] Picked data-all file: data-all-06-19-26-13-39-16-28 (tz_offset=-420min) -> event_utc=2026-06-20 02:34:43

2026-08-12 14:29:20,176 [INFO] Running anb2ann_v4_utc0.exe ...

2026-08-12 14:29:20,190 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\629209_2026-06-19_193443\all_hourly_event.txt

2026-08-12 14:29:20,209 [INFO] Device HW ver: 3.1 FW ver: 4.0.014.3

2026-08-12 14:29:20,267 [INFO] Done. Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\629209_2026-06-19_193443

2026-08-12 14:29:20,300 [INFO] HW ver: 3.1 FW ver: 4.0.014.3

2026-08-12 14:29:20,300 [INFO] StartStudy @ 2026-06-19T20:15:29 UTC0 (from log-daily-06-18-26-17-00-00-28.txt)

FriendlyId = 629209

AlgLead = 2 (middle channel)

Brady/Tachy = 40/145 bpm

PauseThresh = 2500 ms (2.5s)

PauseDetect = enabled

AFibDetect = enabled

Pre/PostEvSecs = 30s / 30s

2026-08-12 14:29:20,301 [INFO] === Automated assessment: Pause at 2026-06-20 02:34:43 UTC0 (window 2026-06-20 02:33:43 .. 2026-06-20 02:35:43) ===

No beats AT ALL were found in this window -- an RR gap couldn't even be computed. See main_cause() for whether a log signal explains this.

=> HES detected ZERO beats near this time -- the ECG acquisition module was repeatedly restarting (log: 260620:023428 ECGA Restarted (0 beat)' at 2026-06-20 02:34:28), consistent with a signal/lead-off problem rather than a genuinely quiet rhythm -> this event was missed because no beat data exists to evaluate, not because of a threshold/consecutive-beat shortfall.

2026-08-12 14:29:20,301 [REPLY] ----- Customer reply draft -----

Study 629209, event time 6/19 19:34:43

1. General information:

FW version: 4.0.014.3, HW version: 3.1

Study's AlgLead is the middle channel (channel 2), not changed during the study

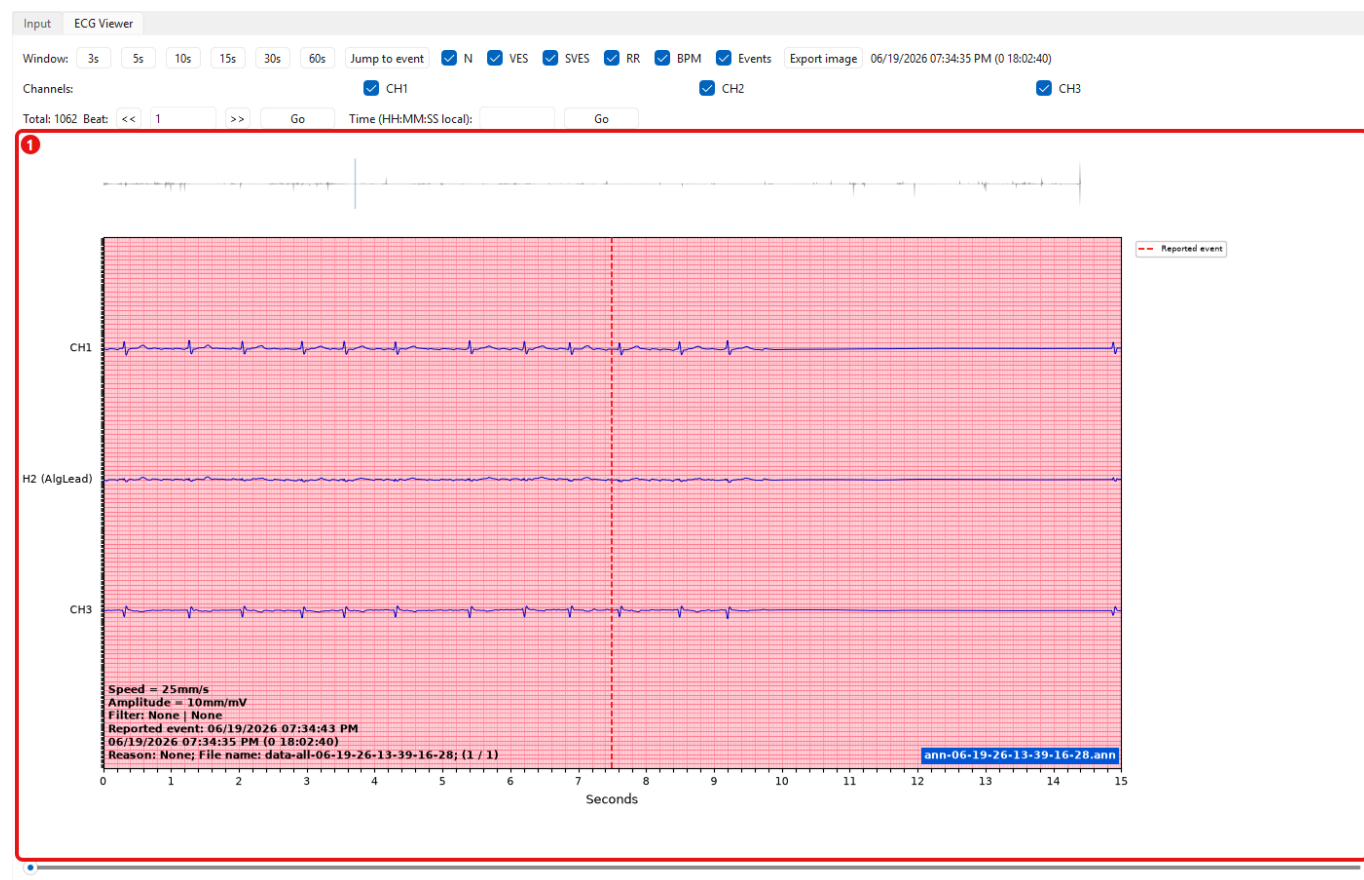
Pause threshold is 2.5s (2500ms)

2. Reason for the missed event:

Main cause: HES detected ZERO beats near this time -- the ECG acquisition module was repeatedly restarting (log: 260620:023428 ECGA Restarted (0 beat)' at 2026-06-20 02:34:28), consistent with a signal/lead-off problem rather than a genuinely quiet rhythm -> this event was missed because no beat data exists to evaluate, not because of a threshold/consecutive-beat shortfall.

1. Progress bar at 100%
2. Status: "Done."
3. Full log and conclusion

Step 4 — ECG Viewer



1. Waveform centered on the reported event — notice the trace goes completely flat right after the dashed "Reported event" line

Conclusion: zero beats at all were found in the $\pm 60s$ window around the reported time — not "beats detected normally with no long-enough gap" as first assumed (see CLAUDE.md's zero-beats bug note). The device log shows the ECG acquisition module repeatedly restarting (**ECGA Restarted (0 beat)**), every ~30-100s for ~18 straight hours spanning this event), consistent with a signal/lead-off problem. This event was missed because there was no beat data to evaluate at all, not because of a threshold shortfall.

Example 4 — Study 611744, Tachy/VT @ 2026-05-21 ~20:51:46

Email: "Can you tell me why the VT episode of 5/21 20:51 did not trigger please?" ("VT" = Ventricular Tachycardia — same event category as "Tachy" in this tool/the device log, just the clinical name.)

Step 1

The screenshot shows the 'Input' tab of the 'ECG Viewer' application. The form contains the following fields and controls:

- Study FID / hash:** A text input field containing '1744', highlighted with a red box and a red circle with the number '1'.
- Input folder:** A text input field with a placeholder '(optional -- reuses this folder if it already has the data, else downloads into it)' and a 'Browse...' button.
- Event type:** A dropdown menu showing '2.chy', highlighted with a red box and a red circle with the number '2'.
- Event date/time (as reported, no conversion needed):** A date and time input field showing '26-05-21' and '20:51:46', highlighted with a red box and a red circle with the number '3'.
- Detected timezone:** A text input field showing '(detected after Run)'.
- Window (+/- sec):** A text input field showing '60'.
- Run button:** A button labeled 'Run', highlighted with a red box and a red circle with the number '4'.
- Open log folder:** A button next to the 'Run' button.
- Progress bar:** A progress bar at the bottom right showing '0%'.
- Result:** A large empty rectangular area for displaying results.

1. Study FID entered
2. Event type selected
3. Event date/time entered exactly as reported (local/study clock)
4. **Run** clicked

Step 2 — mid-run

InputECG Viewer

Study FID / hash:

611744

Input folder:

(optional -- reuses this folder if it already has the data, else downloads into it)

Browse...

Event type:

Tachy

Event date/time (as reported, no conversion needed):

2026-05-21

20:51:46

Detected timezone:

(detected after Run)

Window (+/- sec):

60

Run

Open log folder

1

95%

2

Running...

Result:

3

2026-08-12 14:29:22,537 [INFO] Resolving study hash via get_study_information.py (issue 524)...
2026-08-12 14:29:22,537 [INFO] \$ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_611744.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_611744.json
2026-08-12 14:29:52,838 [INFO] study_hash = 6a05e01d523f149435c7943b
2026-08-12 14:29:54,344 [INFO] Picked data-all file: data-all-05-21-26-17-40-47-16 (tz_offset=-240min) -> event_utc=2026-05-22 00:51:46
2026-08-12 14:30:24,768 [INFO] Running anb2ann_v4_utc0.exe ...
2026-08-12 14:30:24,867 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\611744_2026-05-21_205146\all_hourly_event.txt
2026-08-12 14:30:24,888 [INFO] Device HW ver: 2.1 FW ver: 1.0.0.40

- 1. Progress bar mid-run
- 2. Status: "Running..."
- 3. Live log of the real SSH/S3 steps

Step 3 — final result

InputECG Viewer

Study FID / hash:

611744

Input folder:

(optional -- reuses this folder if it already has the data, else downloads into it)

Browse...

Event type:

Tachy

Event date/time (as reported, no conversion needed):

2026-05-21

20:51:46

Detected timezone:

UTC-4 (detected from data-all-05-21-26-17-40-47-16)

Window (+/- sec):

60

Run

Open log folder

1

100%

2

ie. Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\611744_2026-05-21_205146

Result:

3

2026-08-12 14:29:22,537 [INFO] Resolving study hash via get_study_information.py (issue 524)...
2026-08-12 14:29:22,537 [INFO] \$ W:\report\TOOL_MISSING_EVENT\check-missing-event\venv\Scripts\python.exe W:\report\TOOL_MISSING_EVENT\get-study-information\get_study_information.py --input W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_611744.csv --output W:\report\TOOL_MISSING_EVENT\check-missing-event\tmp_fid_611744.json
2026-08-12 14:29:52,838 [INFO] study_hash = 6a05e01d523f149435c7943b
2026-08-12 14:29:54,344 [INFO] Picked data-all file: data-all-05-21-26-17-40-47-16 (tz_offset=-240min) -> event_utc=2026-05-22 00:51:46
2026-08-12 14:30:24,768 [INFO] Running anb2ann_v4_utc0.exe ...
2026-08-12 14:30:24,867 [INFO] Aggregated hourly events -> W:\report\TOOL_MISSING_EVENT\check-missing-event\output\611744_2026-05-21_205146\all_hourly_event.txt
2026-08-12 14:30:25,140 [INFO] Done. Output folder: W:\report\TOOL_MISSING_EVENT\check-missing-event\output\611744_2026-05-21_205146
2026-08-12 14:30:25,180 [INFO] Device HW ver: 2.1 FW ver: 1.0.0.40
2026-08-12 14:30:25,180 [INFO] StartStudy @ 2026-05-14T17:38:12 UTC0 (from log-daily-05-13-26-20-00-00-16.txt)
FriendllyId = 611744
AlgLead = 2 (middle channel)
Brady/Tachy = 40/145 bpm
PauseThresh = 2500 ms (2.5s)
PauseDetect = enabled
AFibDetect = enabled
Pre/PostEvSecs = 30s / 30s
2026-08-12 14:30:25,180 [INFO] === Automated assessment: Tachy at 2026-05-22 00:51:46 UTC0 (window 2026-05-22 00:50:46 .. 2026-05-22 00:52:46) ===
152 beat(s) found in this window, but none qualifying (HesEventStatus2) near this time.
=> HES detected too few beats -> not enough beats for 5 consecutive beats -> this Tachy event was missed.
2026-08-12 14:30:25,181 [REPLY] ----- Customer reply draft -----
Study 611744, event time 5/21 20:51:46
1. General information:
FW version: 1.0.0.40, HW version: 2.1
Study's AlgLead is the middle channel (channel 2), not changed during the study
Tachy threshold is 145bpm
2. Reason for the missed event:
Main cause: HES detected too few beats -> not enough beats exceeding 145bpm for 5 consecutive beats -> this Tachy event was missed.

1. Progress bar at 100%
2. Status: "Done."
3. Full log and conclusion

Step 4 — ECG Viewer



1. Waveform centered on the reported event

Conclusion: 152 beats were found in the window, but none qualified as Tachy ([HesEventStatus2](#)) — consistent with the answer already given for this case ("HES detected missed beats, heart rate stayed below the 145bpm threshold"). Unlike Example 3, this is a genuine "beats present but rate too low" situation, not a beat-detection dropout.

CLI usage reference

For scripted/automated use (another tool or AI reading its output) instead of the GUI. Entry point:

`src\main.py`.

```
.venv\Scripts\python.exe src\main.py --event-type {Brady,Tachy,Pause} --date YYYY-MM-DD --time HH:MM:SS (--fid FID | --hash HASH | --local-dir DIR) [options]
```

Flags

Flag	Required	Description
<code>--fid</code>	one of <code>--fid/--hash/--local-dir</code>	Study friendly ID (6 digits) — looked up via <code>get_study_information.py</code> (issue 524, real SSH)
<code>--hash</code>		Study hash directly — skips the FID lookup entirely
<code>--local-dir</code>		A folder to reuse (if it already has <code>data-all-*.dat</code>) or download into (if empty/new) — see "double duty" note below
<code>--event-type</code>	yes	Brady, Tachy, or Pause
<code>--date</code>	yes	Event date, local/study clock, <code>YYYY-MM-DD</code>
<code>--time</code>	yes	Event time, local/study clock, <code>HH:MM:SS</code>
<code>--outdir</code>		Output folder (default: <code>output/<fid_or_hash>_<date>_<time>/</code>)
<code>--window-sec</code>		$\pm$ seconds around the event to analyze/plot (default 60)
<code>--dry-run</code>		Print the hash-resolution command without running it or anything after it
<code>--no-plot</code>		Skip generating <code>event_plot_overview.png/event_plot_closeup.png</code>

Exactly one of `--fid`, `--hash`, `--local-dir` is required.

Examples

```
# Real case, FID lookup + S3 download
.venv\Scripts\python.exe src\main.py --fid 629209 --event-type Pause --date 2026-06-19 --time 19:34:43

# Already know the hash? Skip the FID lookup (issue 524 is slow, don't hammer it per email)
.venv\Scripts\python.exe src\main.py --hash <study-hash> --event-type Tachy --date 2026-05-09 --time 17:10:31

# Offline demo/test, or re-opening a case you already downloaded (no AWS needed if data's there)
.venv\Scripts\python.exe src\main.py --local-dir "output\609493_2026-05-09_171031" --event-type Tachy --date 2026-05-09 --time 17:10:31

# Wider  $\pm 120$ s window, custom output folder, skip plotting
.venv\Scripts\python.exe src\main.py --fid 611744 --event-type Tachy --date 2026-05-21 --time 20:51:46 --window-sec 120 --outdir "C:\cases\611744" --no-plot

# Just see the resolved hash without downloading anything
```



```
.venv\Scripts\python.exe src\main.py --fid 609493 --event-type Tachy --date 2026-05-09 --time 17:10:31 --dry-run
```

Notes

- `--date/--time` are the event timestamp **exactly as it appears in the recording site's own local clock** — the same clock used by the `data-all/report-hourly` file names and `all_hourly_event.txt`. Resolve any AM/PM or day/month ambiguity yourself before entering it; the tool does not guess.
- The tool derives the UTC0 offset for you from the tzcode embedded in the downloaded file names — you never need to pass a timezone.
- `--dry-run` only prints the hash-resolution command; the rest of the pipeline needs the real hash to know what to list/download, so nothing past that point is covered by dry-run.
- `--local-dir` serves double duty, decided by what's already in it: it **reuses** the folder if it already has `data-all-*.dat` files (skips hash resolution + all S3 downloads entirely, even if `--fid/--hash` is also given), or, if empty/new, uses it as the **download destination** instead of the default `output/<fid_or_hash>_<date>_<time>/` (requires `--fid/--hash` in that case).
- On success, prints the parsed study settings (if found) and the automated assessment to stdout; on an expected failure (e.g. no `data-all/` object, `PauseThresh` unknown), prints a clear one-line error and exits 1 rather than a raw traceback.